

Reproduce-then-Extend — Pinyon Jay Habitat Selection (Ecology)

(Day 2)

Intro to Machine Learning · Bruce A. Desmarais

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Published study. Daw, S. K. et al. (2022). “Behavior-specific occurrence patterns of Pinyon Jays.” *PLoS ONE* (doi:10.1371/journal.pone.0237621; S1 Dataset, open). A **logistic resource-selection function (RSF)** — habitat *use* points (1) vs *available* points (0) on environmental predictors. Use is the minority class, so out-of-sample skill is scored by **AUC-PR** (area under the precision-recall curve).

1 Background

Pinyon Jays are a **declining bird** of the southwestern United States, and managers need to know **which habitat they use** in order to conserve it. Daw and colleagues fit a **resource-selection function** – a logistic regression comparing locations the jays *used* to *available* background locations – to identify the environmental conditions associated with jay occurrence. The **primary conclusion** is that jays select particular pinyon-juniper conditions – more open stands, sites near habitat edges, and areas with downed woody debris – which the regression translates into a picture of suitable habitat for management.

2 Data and codebook

Unit of analysis: a sampled ground location – an observed jay-**use** point vs an **available** (background) point; $n = 466$ (124 use).

Variable	Definition
use	1 = observed Pinyon Jay use location, 0 = available location (outcome)
Elevation..feet.	elevation (feet)
Slope..percent.	ground slope (%)
Stand_Age..years.	forest stand age (years)
Stand_Density_Index	stand density index (tree density / competition)
Canopy_Cover..percent.	overstory canopy cover (%)
Tree_Cover..percent.	tree cover (%)
Shrub_Cover..percent.	shrub cover (%)
Forb_Cover..percent.	forb (herbaceous) cover (%)
Grass_Cover..percent.	grass cover (%)
Woody_Debris..category.1.-.3.	counts of downed woody debris by size class (1-3)
Distance.to.Road..km.	distance to the nearest road (km)
DisturbanceYes	1 if any recent disturbance was recorded at the site
Distance.to.Edge..m.	distance to the nearest habitat edge (m)

3 Descriptive analysis

```
d <- read.csv(paste0(
  "https://raw.githubusercontent.com/bdesmarais/intro",
  "-ml-statistical-horizons-4day/main/tutorials/day4_",
  "classic_learners/06_pinyonjay_rsf/data/jays_clean.",
  "csv"))

# Predictors: keep the interpretable environmental variables and drop the sparse
# one-hot habitat/disturbance-type dummies (unidentified in the logit).
preds <- setdiff(names(d), "use")
preds <- preds[sapply(d[preds], function(x) length(unique(x)) > 1)]
preds <- preds[!grepl("Habitat_Type|Disturbance_Type", preds)]

clean <- function(v) gsub("[. _]+", " ", v) # tidy mangled names for labels
short <- function(v) { # compact labels for tables/axes
  v <- gsub("[. _]+", " ", v); v <- gsub("percent", "%", v)
  v <- gsub("category", "cat", v); trimws(v)
}
num <- setdiff(preds, "DisturbanceYes") # 14 numeric environmental vars
```

3.1 Dataset and provenance

The unit of analysis is a **sampled ground location** in pinyon–juniper woodland of the southwestern United States. Each row is either a location where Pinyon Jays were **observed** (a *use* point) or a randomly drawn **available** (background) point that characterizes what habitat was on offer. This use-vs-available design is the standard setup for a **resource-selection function**: the model learns which environmental conditions distinguish used from available locations. The data are the open **S1 Dataset** from Daw et al. (2022, *PLoS ONE*).

There are **466 locations** (124 use, 342 available) and **15 predictors** retained for modeling — 14 continuous environmental measurements plus one binary disturbance indicator. (The raw file also carries sparse one-hot `Habitat_Type*` and `Disturbance_Type*` dummies; these are dropped above because they are near-collinear and unidentified in the logistic RSF.)

Table 1: Variable dictionary: outcome and retained predictors.

Variable	Role	Type	Description
use	outcome	binary	1 = jay use location, 0 = available
Elevation..feet.	predictor	numeric	elevation (feet)
Slope..percent.	predictor	numeric	ground slope (%)
Stand_Age..years.	predictor	numeric	forest stand age (years)
Stand_Density_Index	predictor	numeric	stand density index (competition)
Canopy_Cover..percent.	predictor	numeric	overstory canopy cover (%)
Tree_Cover..percent.	predictor	numeric	tree cover (%)
Shrub_Cover..percent.	predictor	numeric	shrub cover (%)
Forb_Cover..percent.	predictor	numeric	forb (herb) cover (%)
Grass_Cover..percent.	predictor	numeric	grass cover (%)
Woody_Debris..category.1.	predictor	count	downed wood, small size class
Woody_Debris..category.2.	predictor	count	downed wood, medium size class
Woody_Debris..category.3.	predictor	count	downed wood, large size class
Distance.to.Road..km.	predictor	numeric	distance to nearest road (km)
DisturbanceYes	predictor	binary	1 if recent disturbance recorded
Distance.to.Edge..m.	predictor	numeric	distance to nearest habitat edge (m)

Variable	Role	Type	Description
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3.2 The outcome: used vs available

The outcome **use** is **binary and imbalanced**: only **26.6%** of locations are use points, so *available* is the majority class at a **2.8:1** ratio. This base rate is the single most important fact for evaluation. A trivial model that always predicts “available” would already be 73.4% accurate, so **raw accuracy is misleading** here — a good classifier must find the minority *use* class. That is exactly why the tutorial scores held-out skill with **AUC-PR** (area under the precision–recall curve), whose no-skill baseline equals the base rate (0.266), not 0.5.

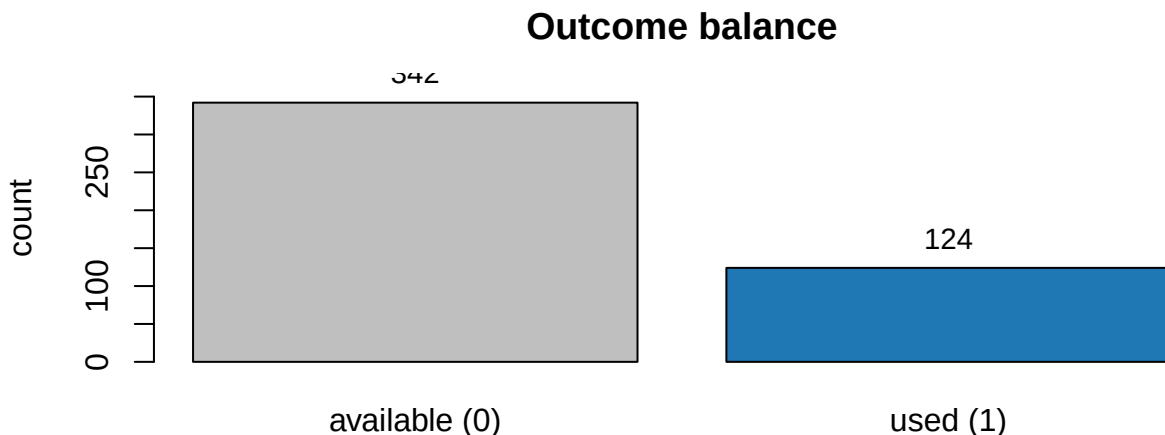


Figure 1: Class balance of the use-vs-available outcome. Use is the minority class.

3.3 Univariate predictor distributions

The 14 continuous predictors span very different scales (elevation in thousands of feet, cover in percent, debris as small counts), so we summarize them in a single table with location, spread, quartiles and **skewness**, then show the shapes as faceted histograms.

Table 2: Univariate summary of the numeric predictors.

	n	miss	mean	sd	min	Q1	med	Q3	max	skew
Elevation feet	466	0	6810.0	686.1	4430	6275.5	6760.0	7270.0	8700.0	0.2
Slope %	466	0	28.4	17.6	0	15.0	25.0	40.0	91.0	0.7
Stand Age years	466	0	143.2	81.2	0	86.0	136.0	194.8	548.0	0.6
Stand Density Index	466	0	154.7	97.5	0	82.7	146.8	220.7	542.0	0.5
Canopy Cover %	466	0	27.9	16.0	0	17.0	28.0	40.0	74.0	0.1
Tree Cover %	466	0	19.1	13.4	0	7.8	18.8	28.6	59.8	0.4
Shrub Cover %	466	0	6.1	6.0	0	2.0	4.0	7.9	37.2	1.9
Forb Cover %	466	0	1.6	2.6	0	1.0	1.0	1.2	26.0	5.9
Grass Cover %	466	0	5.8	8.8	0	1.2	2.2	5.8	59.8	3.1
Woody Debris cat 1	466	0	15.3	21.4	0	2.0	6.0	19.0	180.0	2.7
Woody Debris cat 2	466	0	2.6	4.8	0	0.0	1.0	3.0	46.0	4.5
Woody Debris cat 3	466	0	0.9	2.2	0	0.0	0.0	1.0	20.0	4.4
Distance to Road km	466	0	6.0	1.6	1	5.0	6.0	7.0	9.0	-0.9

	n	miss	mean	sd	min	Q1	med	Q3	max	skew
Distance to Edge m	466	0	1684.2	2015.4	-676	315.2	1000.5	2299.2	10800.0	1.7

Several patterns stand out. The **cover** and **woody-debris** variables are strongly **right-skewed** (skew for forb, grass and the debris counts all exceed 2): most locations have little cover or downed wood, with a long tail of rich sites. **Elevation** and **canopy cover** are close to symmetric. **Distance.to.Edge.m.** reaches a large maximum (1.08×10^4 m) with a long right tail, and it contains a few **negative values** (min = -676), an artifact of how signed distance-to-edge was computed in the source GIS layer — worth noting but harmless for tree-based learners, which respond only to rank order.

The one binary predictor, **DisturbanceYes**, is recorded at 13.3% of locations (62 of 466) — a minority flag for sites with any recorded recent disturbance.

3.4 Missing data

Total missing cells across outcome + predictors: 0

Complete cases: 466 of 466 (100%)

The analysis matrix is **fully observed** — every outcome and predictor cell is present, so there is no missingness to model and no co-missingness pattern to inspect. The published RSF therefore uses all 466 locations with no imputation or listwise deletion, and the machine-learning comparison inherits the same complete design.

3.5 Bivariate association with the outcome

We next ask which conditions actually separate used from available locations. For each numeric predictor we report the **point-biserial correlation** with **use**, the **standardized mean difference** (used minus available, in SDs), and a two-sample *t*-test *p*-value; the table is sorted by the magnitude of the standardized difference.

Table 3: Marginal association of each numeric predictor with use, sorted by |std. mean difference|.

	r	smd	p
Woody Debris cat 1	0.422	0.955	<1e-04
Distance to Edge m	-0.415	-0.938	<1e-04
Woody Debris cat 2	0.414	0.935	<1e-04
Distance to Road km	-0.394	-0.891	<1e-04
Slope %	-0.381	-0.862	<1e-04
Woody Debris cat 3	0.320	0.723	<1e-04
Stand Density Index	-0.316	-0.714	<1e-04
Elevation feet	-0.312	-0.706	<1e-04
Grass Cover %	0.231	0.522	<1e-04
Stand Age years	-0.204	-0.462	<1e-04
Canopy Cover %	-0.146	-0.331	0.0012
Forb Cover %	0.088	0.199	0.1647
Tree Cover %	-0.026	-0.059	0.5476
Shrub Cover %	0.011	0.025	0.7980

The signal is clear and ecologically sensible. Use rises with **downed woody debris** (all three size classes positive), and falls sharply with **distance to edge**, **distance to road**, **slope**, **stand density** and **elevation** — jays select **open, low-density stands near edges with downed wood**, and avoid steep, dense, high-elevation interior sites. Tree and shrub cover show essentially **no marginal association** ($|r| < 0.03$), a hint that their role, if any, is nonlinear or conditional rather than a simple linear gradient. For the binary flag:

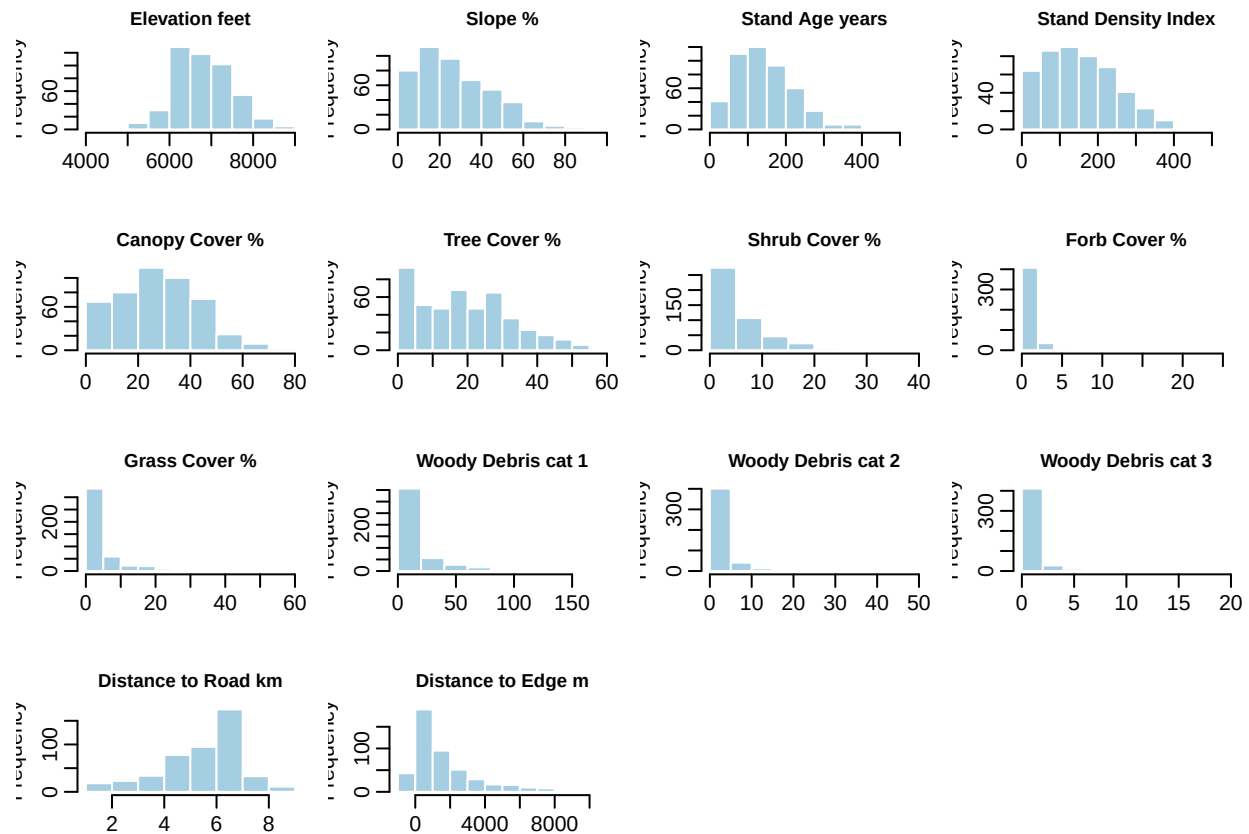


Figure 2: Distributions of the numeric predictors. Cover and woody-debris variables are right-skewed; elevation and canopy are roughly symmetric.

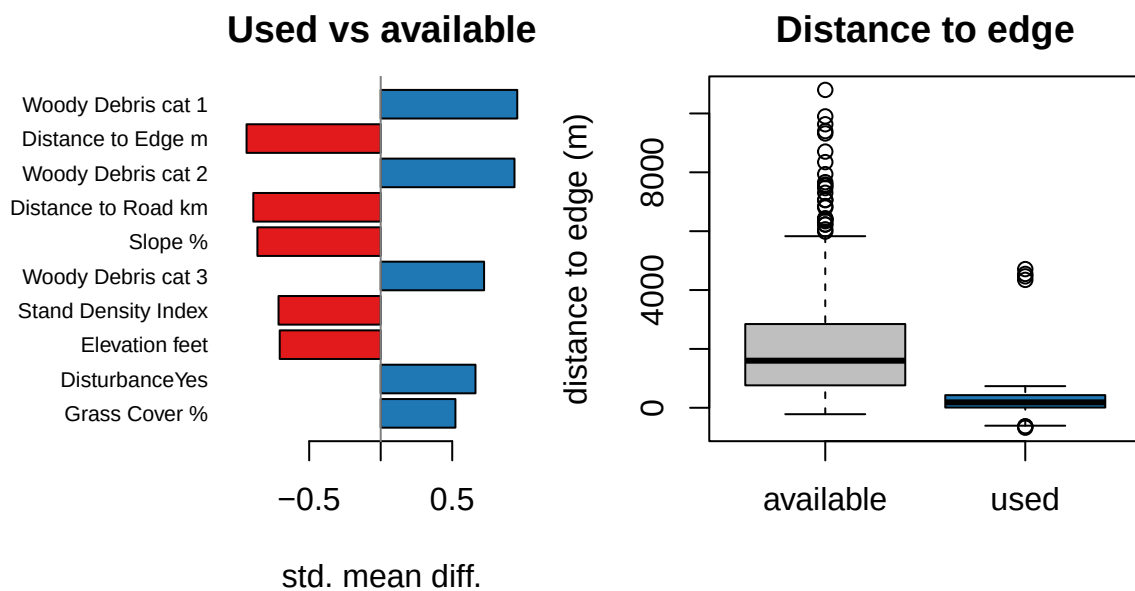


Figure 3: Standardized mean difference (used - available) for each predictor (left) and the strongest gradient, distance-to-edge, by outcome (right).

```
## Use rate by disturbance flag:
```

```
##      0      1
## 0.215 0.597
```

```
## chi-square test of independence: p = <1e-04
```

Disturbed sites are used far more often (60% vs 22%), a strong and highly significant association.

3.6 Multivariate structure and collinearity

Finally we check whether the predictors are redundant with one another. The heatmap shows pairwise correlations among the numeric predictors, and we compute **variance-inflation factors** from the reproduction model (via auxiliary regressions, since a linear RSF shares the design matrix).

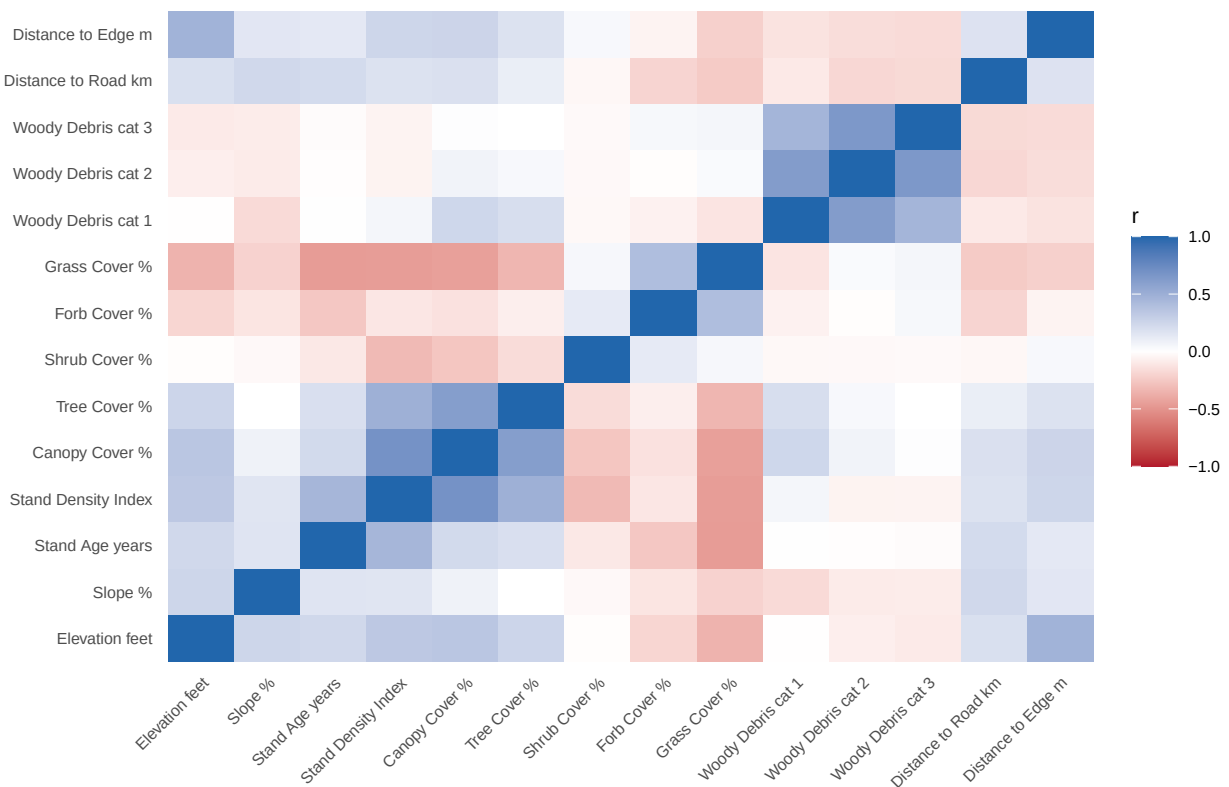


Figure 4: Correlation heatmap of the numeric predictors. No pair exceeds $|r| = 0.7$; the strongest is canopy vs stand-density index.

```
## Largest off-diagonal |r| among numeric predictors: 0.69
```

```
## Variance-inflation factors (top 5):
```

```
##      Canopy_Cover..percent.      Stand_Density_Index Woody_Debris..category.2.
##              2.78              2.61              2.35
## Woody_Debris..category.1.      Grass_Cover..percent.
##              1.96              1.86
```

Collinearity is **mild**: the strongest correlation is only 0.69 (canopy cover vs stand-density index — both overstory-structure measures), **no pair exceeds** $|r| = 0.7$, and the **maximum VIF is 2.78**, far below the usual threshold of 5–10. The predictors therefore carry largely distinct information; the coefficients in the reproduction are individually interpretable, and the modest correlations that do exist (structure, debris size classes) are the kind of interaction-rich texture the tree-based learners can exploit later.

4 Reproduce the published RSF regression

The published model is a logistic regression of use vs availability on the environmental predictors.

```
rsf <- glm(reformulate(preds, "use"), data = d, family = binomial)
round(summary(rsf)$coefficients, 3)
```

	Estimate	Std. Error	z value	Pr(> z)
## (Intercept)	6.735	2.827	2.383	0.017
## Elevation..feet.	0.000	0.000	-1.080	0.280
## Slope..percent.	-0.067	0.015	-4.495	0.000
## Stand_Age..years.	0.001	0.003	0.322	0.748
## Stand_Density_Index	-0.024	0.005	-4.699	0.000
## Canopy_Cover..percent.	0.034	0.023	1.500	0.134
## Tree_Cover..percent.	0.080	0.026	3.108	0.002
## Shrub_Cover..percent.	-0.015	0.035	-0.435	0.664
## Forb_Cover..percent.	-0.056	0.084	-0.668	0.504
## Grass_Cover..percent.	-0.039	0.027	-1.464	0.143
## Woody_Debris..category.1.	0.029	0.014	2.132	0.033
## Woody_Debris..category.2.	0.197	0.076	2.597	0.009
## Woody_Debris..category.3.	0.005	0.105	0.047	0.962
## Distance.to.Road..km.	-0.452	0.134	-3.383	0.001
## DisturbanceYes	2.269	0.559	4.055	0.000
## Distance.to.Edge..m.	-0.001	0.000	-4.763	0.000

Confirmation against the published study. The signed coefficients recover the reported habitat-selection pattern — jays select more **open** stands (negative on `Stand_Density_Index` and `Slope..percent.`) and sites **closer to edges** (negative on `Distance.to.Edge..m.`). This logistic RSF is our regression baseline.

5 Predictive results: regression vs machine learning

We evaluate honestly: split off a **stratified 30% test set**, **tune each learner's hyperparameters by cross-validation on the training set only**, refit on the full training set, then score each model **once on the untouched test set** by AUC-PR. Nothing about the test set is used for fitting or tuning.

```
X <- as.matrix(d[, preds]); y <- d$use
aucpr <- function(p, yt) pr.curve(scores.class0 = p[yt == 1], scores.class1 = p[yt ==
  ↪ 0])$auc.integral

set.seed(2026)
te <- c(sample(which(y == 1), round(0.30 * sum(y == 1))),      # stratified 30% test set
        sample(which(y == 0), round(0.30 * sum(y == 0))))
tr <- setdiff(seq_len(nrow(d)), te); yte <- y[te]

## 1. Logistic RSF -- the published regression (no hyperparameters)
m_logit <- glm(use ~ ., data = d[tr, c("use", preds)], family = binomial)
test_auc <- c(Logit = aucpr(predict(m_logit, d[te, ], type = "response"), yte))

## 2. Elastic net -- penalty lambda tuned by CV on the TRAINING set
m_en <- cv.glmnet(X[tr, ], y[tr], alpha = 0.5, family = "binomial")
test_auc["ElasticNet"] <- aucpr(as.numeric(predict(m_en, X[te, ], s = "lambda.min", type
  ↪ = "response")), yte)

## 3. Random forest -- mtry tuned on the TRAINING set by out-of-bag AUC-PR
best_rf <- NULL
for (mt in unique(pmax(1, c(floor(sqrt(ncol(X))), floor(ncol(X) / 3), floor(ncol(X) /
  ↪ 2)))) {
```

```

rf <- ranger(x = X[tr, ], y = factor(y[tr]), num.trees = 800, mtry = mt, probability =
→ TRUE)
sc <- aucpr(rf$predictions[, "1"], y[tr]) # OOB predictions = a training-only
→ CV estimate
if (is.null(best_rf) || sc > best_rf$sc) best_rf <- list(mtry = mt, sc = sc)
}
m_rf <- ranger(x = X[tr, ], y = factor(y[tr]), num.trees = 800, mtry = best_rf$mtry,
→ probability = TRUE)
test_auc["RandomForest"] <- aucpr(predict(m_rf, X[te, ])$predictions[, "1"], yte)

## 4. XGBoost -- depth x eta tuned by xgb.cv on the TRAINING set; nrounds by early
→ stopping
dtrain <- xgb.DMatrix(X[tr, ], label = y[tr])
best <- NULL
for (depth in c(3, 4, 6)) for (eta in c(0.05, 0.1)) {
  xcv <- xgb.cv(params = list(objective = "binary:logistic", max_depth = depth, eta =
→ eta,
                                subsample = 0.9, colsample_bytree = 0.8, eval_metric =
→ "aucpr"),
                data = dtrain, nrounds = 600, nfold = 5,
                early_stopping_rounds = 25, verbose = 0, maximize = TRUE)
  score <- max(xcv$evaluation_log$test_aucpr_mean)
  nr <- if (!is.null(xcv$best_iteration)) xcv$best_iteration else
→ nrow(xcv$evaluation_log)
  if (is.null(best) || score > best$score) best <- list(depth = depth, eta = eta, nrounds
→ = nr, score = score)
}
m_xgb <- xgb.train(params = list(objective = "binary:logistic", max_depth = best$depth,
→ eta = best$eta,
                                subsample = 0.9, colsample_bytree = 0.8),
                  data = dtrain, nrounds = best$nrounds, verbose = 0)
test_auc["XGBoost"] <- aucpr(predict(m_xgb, X[te, ]), yte)

round(test_auc, 3)

```

```

##      Logit   ElasticNet RandomForest      XGBoost
##      0.953      0.970      0.969      0.970

```

```

cat(sprintf("Best ML gain over the logistic RSF on the held-out test set: +%.3f
→ AUC-PR\n",
            max(test_auc[-1]) - test_auc["Logit"]))

```

```
## Best ML gain over the logistic RSF on the held-out test set: +0.018 AUC-PR
```

On the held-out test set, random forest and XGBoost lift AUC-PR over the logistic RSF — the nonlinear, interaction-rich structure of habitat selection that a linear-in-the-logit model cannot capture.

6 Interpreting the model: importance, SHAP, and partial dependence

We open the fitted learners three ways: **permutation importance**, **SHAP values** (exact TreeSHAP — the signed contribution of each variable to each prediction), and **partial dependence** (the shape of an effect).

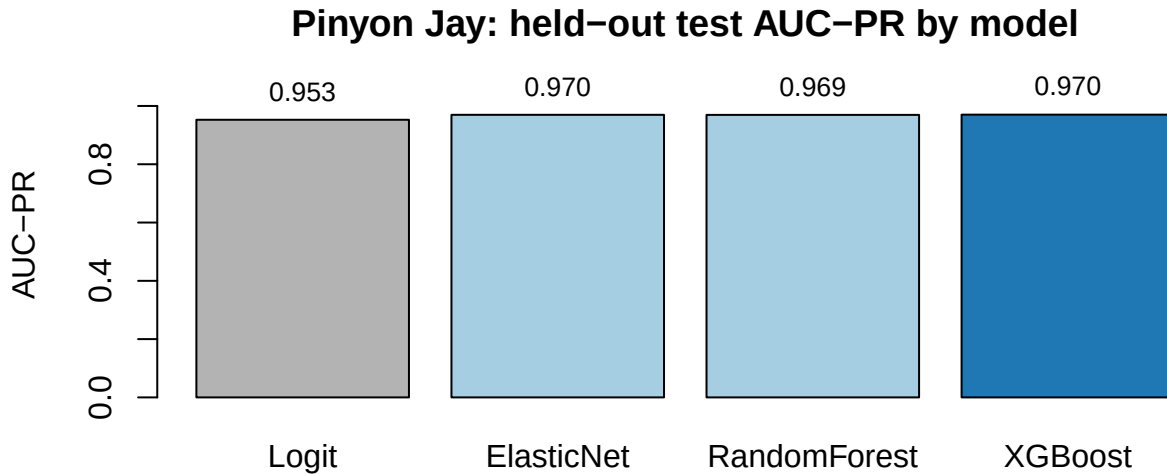


Figure 5: Held-out test-set AUC-PR (higher is better). Grey = published logistic RSF; blue = best ML.

```
# Random forest (permutation importance + partial dependence) on all the data
rf <- ranger(x = X, y = factor(y), num.trees = 800, probability = TRUE, importance =
  ↪ "permutation")

# XGBoost fit on all the data, then exact TreeSHAP contributions per observation
xgb <- xgb.train(params = list(objective = "binary:logistic", max_depth = 4, eta = 0.05,
  subsample = 0.9, colsample_bytree = 0.8),
  data = xgb.DMatrix(X, label = y), nrounds = 300, verbose = 0)
shap <- predict(xgb, X, predcontrib = TRUE) # last column is the model bias
shap <- shap[, colnames(X)]                # drop the BIAS column
```

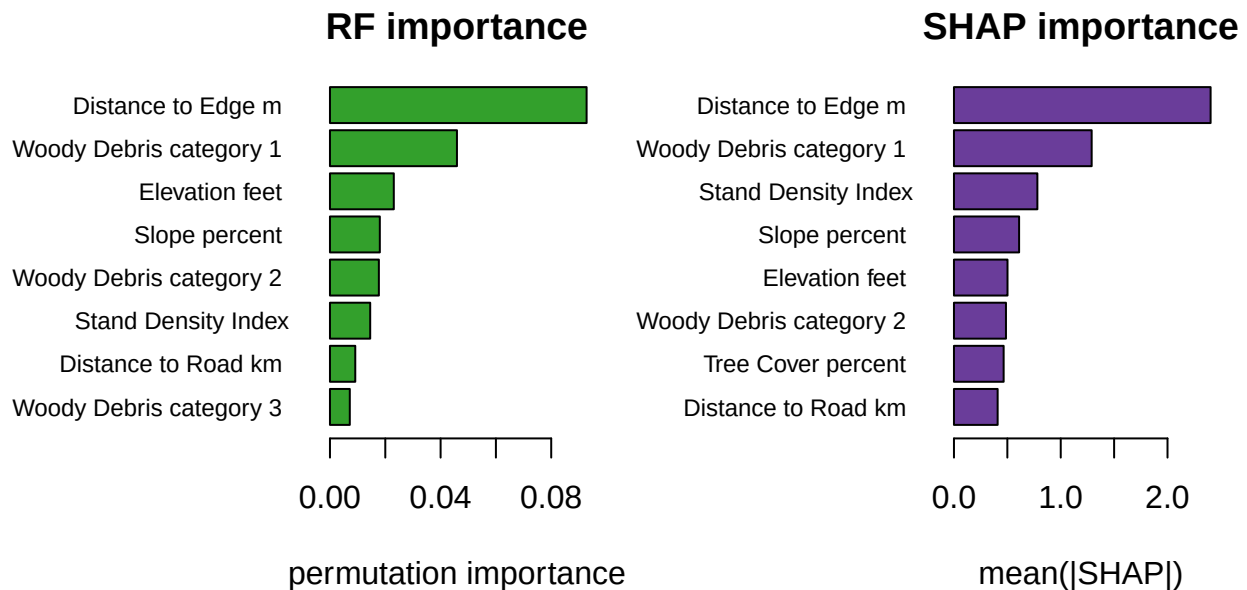


Figure 6: Permutation importance (left) and mean-absolute-SHAP importance (right). The two rankings agree.

Interpreting the relationships. The SHAP summary reads directly as habitat selection: use rises as **stand-density index falls** (open blue points sit at positive SHAP, dense red at negative) and **closer to**

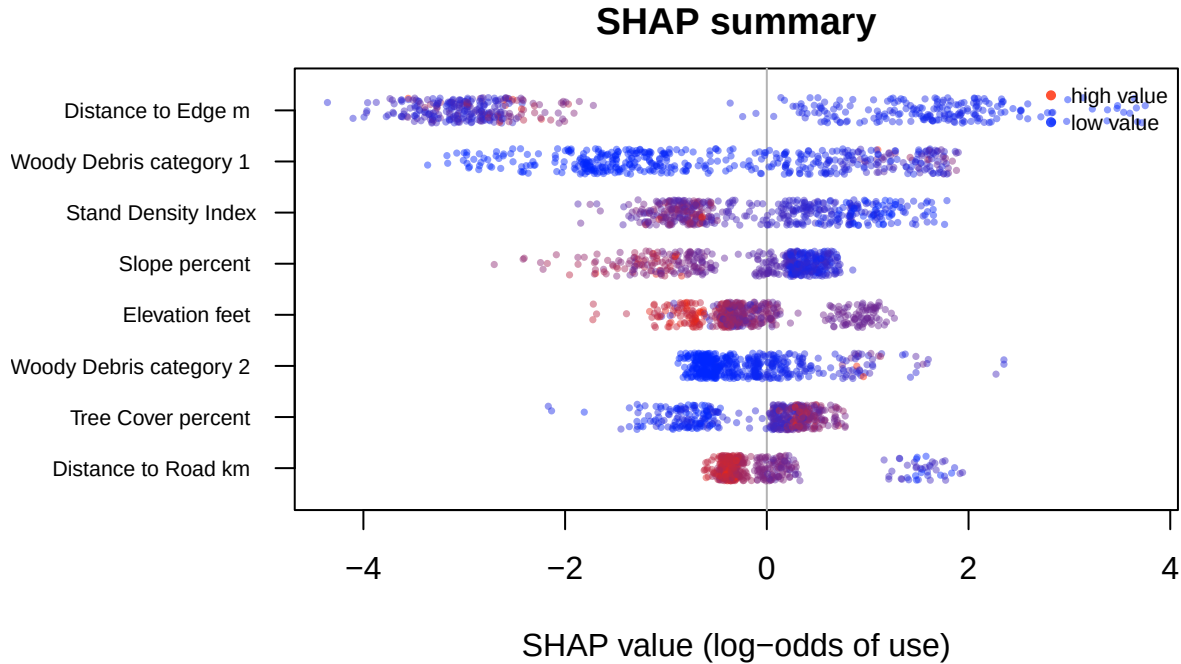


Figure 7: SHAP summary. Each point is one location; x = its SHAP contribution to the log-odds of use; colour = the variable's value (blue low, red high).

edges, with **woody debris** present. In plain terms, pinyon jays select **open pinyon–juniper woodland near edges with downed wood** — and avoid dense, closed stands.

Both curves show a strongly **nonlinear** effect — the contribution turns over across the range rather than following one logit slope, which (with the interactions SHAP captures locally) is why the learners beat the linear RSF.

7 Takeaway

Birds select habitat at nonlinear optima and via interactions — where the linear RSF struggles and trees excel — and SHAP lets us read those relationships (open stands, edges, woody debris) directly off the fitted model. The first of two ecology cases showing ML both predicts better *and* interprets cleanly.

8 Recommended exercises

1. Vary the random-forest hyperparameters (number of trees, features per split) and report the effect on AUC-PR.
2. Generate a partial-dependence plot for a second habitat variable and describe the nonlinearity.
3. Compare RF vs XGBoost AUC-PR on the same folds; which wins, and by how much?

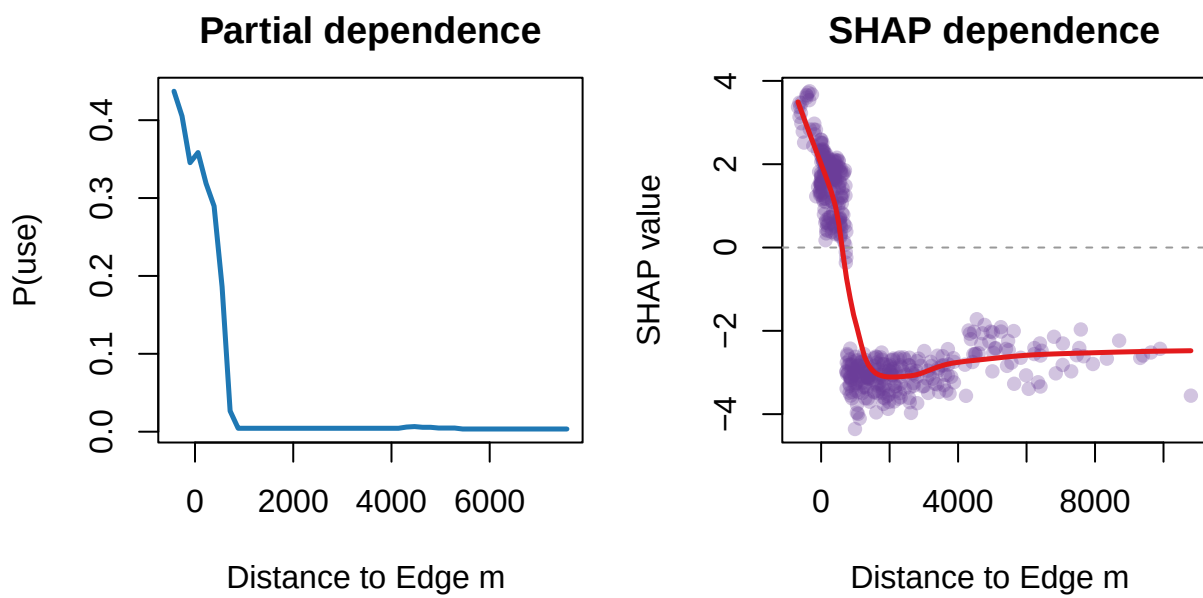


Figure 8: Partial dependence (left) and SHAP dependence (right) for the most influential variable.